Student Registration Number

C

Course Code: DMGT204

Course Title: QUANTITATIVE TECHNIQUES-I

Time Allowed: 3 hours **Max. Marks: 80**

***1**. This paper contains 10 questions divided in two parts on __2__ pages.*

2. Part A is compulsory.

3. In Part B (Questions 2 to 10), attempt any 6 questions out of 9. Attempt all parts of the selected question.

***4.** The marks assigned to each question are shown at the end of each question in square brackets.*

***5.** Answer all questions in serial order.*

6. The student is required to attempt the question paper in English medium only.

PART A

Q1. a) Explain in details relation between sample and population. [2]

b) What is Bar chart? [2]

c) Explain Histogram in few words. [2]

d) Calculate deviation from mean, for the given data: 2, 4, 6, 8, 12, 13, 17. [2]

e) Write down four different methods of collecting primary data. [2]

f) Define harmonic mean. [2]

g) Define kurtosis of a distribution. [2]

h) An unbiased coin is tossed three times. What is the chance of getting three head? [2]

i) Find $P(A \cup B)$ where A and B are two mutually exclusive events and it is given that $P(A) = P(B) = 1/2$. [2]

j) If a normal variate has it's value 2013, what is the correspondence in standard normal variate? [2]

PART B

Q2. Draw the histogram and calculate the arithmetic mean for the following distribution.

Class interval:	0-99	100-199	200-299	300-399	400-499	500-599	600-699	700-799
Frequency:	10	54	184	264	246	40	1	1.

[10]

Q3. Determine mean, median, mode from the following distribution

Class intervals:	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45
Frequencies:	5	6	15	10	5	4	2	2.

[10]

Q4. Compute the quartile deviation and standard deviation for the following:

X:	100-109	110-119	120-129	130-139	140-149	150-159	160-169	170-179
F:	15	44	133	150	125	82	35	16.

[10]

Q5. In a partially destroyed laboratory record, only the line of regression of y on x and x on y as 4x-5y= -33 and 20x-9y=107 respectively calculate the mean of them and correlation coefficient between them. [10]

Q6. The students in a class are selected random, one after another for an examination. Find the probability p that the boys and girls in the class alternate if

(a) The class consists of 4 boys and 3 girls.

(b) The class consists of 3 boys and 3 girls. [10]

Q7. Fit a binomial distribution for the following data and compare the theoretical frequencies with the actual ones:

X :	0	1	2	3	4	5	6
F :	13	25	52	58	32	16	4

[10]

Q8. In an examination taken by 500 candidates, the average and the standard deviation of marks obtained (normally distributed) are 40% and 10%. Find approximately

(a) How many will pass if 50% is fixed as minimum?

(b) What should be the minimum if 350 candidates are to pass?

(c) How many have scored marks above 60%? [10]

Q9. In a test of 2000 electric bulbs, it was found that life of a particular make, was normally distributed with an average life of 2040 hours and S. D. of 60 hours. Estimates the number of bulbs likely to burn for:

(a) More than 2150 hours,

(b) Less than 1950 hours and

(c) More than 1920 hours but less than 2160 hours. [10]

Q10. Find the rank correlation for the following data:

X: 56	42	72	36	63	47	55	49	38	42	68	60
Y: 147	125	160	118	149	128	150	145	115	140	152	155

[10]